

Point

From Number Theory to Algebraic Geometry and Logic

Session P2 — Topology and Metric Spaces

Complete lecture notes with proofs, examples, and exercises

Abstract

This session builds the language of *closeness*. Two machines are constructed and proved correct, because the entire course runs on them: **compactness** (which turns open covers into finite data — the engine behind adelic finiteness in S22 and profinite compactness in S34) and **completion** (which adds exactly the points that Cauchy sequences deserved — the machine that turns $\mathbb{Q}$ into $\mathbb{R}$ here and into $\mathbb{Q}_p$ in S17–S18). Along the way: bases, subspace and product topologies (with the infinite-product remark that previews the restricted product of adeles), connectedness of intervals, the Heine–Borel picture, Hausdorffness, and the equivalence of ε – δ and open-set continuity.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: closeness without distance; where topology returns (S1 Zariski, S17–S22, S34, S38)
0:10–0:35	§1 Topological spaces: axioms, bases, subspace & product, continuity, homeomorphism
0:35–1:00	§1 cont.: connectedness of intervals; compactness of the unit interval; Hausdorff
1:00–1:10	<i>Break</i>
1:10–1:35	§2 Metric spaces: balls as a basis; ε – $\delta \iff$ open-set continuity; Cauchy; completeness
1:35–2:05	§3 Completion: construction, completeness, density, uniqueness; $\mathbb{Q} \rightarrow \mathbb{R}$; preview $\mathbb{Q} \rightarrow \mathbb{Q}_p$, Ostrowski
2:05–2:20	§4 Compactness in metric spaces; uniform continuity; extreme values
2:20–2:30	Wrap-up: completion = adding missing points; bridge to S17–S22

Conventions

All spaces are nonempty unless stated. “Open” always means open in the topology under discussion; subspace and product topologies are defined explicitly when used. In metric contexts d denotes the metric and $B(x, r) = \{y : d(x, y) < r\}$ the open ball. Sequences are written $(x_n)_{n \geq 1}$ or (x_n) .

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1 Topological Spaces

Definition 1.1 (Topology). A **topology** on a set X is a collection $\tau \subseteq \mathcal{P}(X)$ of **open sets** with:

- (T1) $\emptyset \in \tau$ and $X \in \tau$;
- (T2) arbitrary unions of members of τ lie in τ ;
- (T3) finite intersections of members of τ lie in τ .

A set is **closed** if its complement is open. The pair (X, τ) is a **topological space**.

Example 1.2 (First examples). • **Discrete:** $\tau = \mathcal{P}(X)$. Every subset open (and closed).

- **Indiscrete:** $\tau = \{\emptyset, X\}$.
- **Standard on $\mathbb{R}$:** τ generated by open intervals (a, b) (made precise by Proposition 1.4); equivalently the metric topology of $|x - y|$ (§2).
- **Cofinite:** $\tau = \{\emptyset\} \cup \{U : X \setminus U \text{ finite}\}$ on any infinite X (check (T2),(T3): unions of cofinite are cofinite; intersections of finitely many cofinite are cofinite).
- **Zariski:** on $\text{Spec } R$ (Session S1) and on k^n (zero sets of polynomials closed); recall it is *not* Hausdorff — Remark 1.24.

Definition 1.3 (Basis). A **basis** for a topology on X is a collection $\mathcal{B} \subseteq \mathcal{P}(X)$ with:

- (B1) $\bigcup_{B \in \mathcal{B}} B = X$;
- (B2) if $x \in B_1 \cap B_2$ with $B_i \in \mathcal{B}$, then $\exists B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

Proposition 1.4 (A basis generates a topology). *If $\mathcal{B}$ satisfies (B1),(B2), then*

$$\tau = \left\{ U \subseteq X : \forall x \in U \exists B \in \mathcal{B}, x \in B \subseteq U \right\}$$

is a topology, and τ is exactly the set of unions of members of $\mathcal{B}$.

Proof. $\emptyset \in \tau$ vacuously; $X \in \tau$ by (B1). Unions: if $x \in \bigcup_i U_i$ then $x \in U_j$ for some j , so a witness B for U_j witnesses $\bigcup_i U_i$. Intersections: if $x \in U_1 \cap U_2$ with witnesses B_1, B_2 , then (B2) gives B_3 with $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$. Every union of basis elements is in τ (each point witnessed by its own B); conversely each $U \in \tau$ equals $\bigcup \{B \in \mathcal{B} : B \subseteq U\}$ by definition. $\square$

Definition 1.5 (Subspace topology). For $Y \subseteq X$, the **subspace topology** on Y has as opens exactly $\{U \cap Y : U \in \tau_X\}$.

Example 1.6. In $Y = (0, 1] \subseteq \mathbb{R}$ the set $(\frac{1}{2}, 1]$ is *open* (it is $(\frac{1}{2}, 2) \cap Y$) although it is not open in $\mathbb{R}$. In $Y = \mathbb{Z} \subseteq \mathbb{R}$ every subset is open (discrete subspace), since $\{n\} = (n - \frac{1}{2}, n + \frac{1}{2}) \cap \mathbb{Z}$.

Definition 1.7 (Product topology). On $X \times Y$, the **product topology** has basis $\{U \times V : U \in \tau_X, V \in \tau_Y\}$ (the basis axioms hold trivially). For finitely many factors, basis $\prod_i U_i$.

Remark 1.8 (Infinite products, and a preview of adeles). For an infinite family $\prod_i X_i$, the **product topology** takes as basis the sets $\prod_i U_i$ with U_i open and $U_i = X_i$ for *all but finitely many* i ; the **box topology** drops this restriction and is too fine (products of compact spaces fail to be compact, etc.). The “all but finitely many” discipline returns in Session S22: the adèle ring $\mathbb{A}_{\mathbb{Q}}$ is a *restricted* product, $\{(x_v)_v : x_p \in \mathbb{Z}_p \text{ for all but finitely many } p\}$ — the same philosophy of controlled infinity, now arithmetic.

Definition 1.9 (Continuity; homeomorphism). $f : X \rightarrow Y$ is **continuous** if $f^{-1}(V)$ is open for every open $V \subseteq Y$. A **homeomorphism** is a bijective continuous map with continuous inverse; then $X \cong Y$ topologically.

Proposition 1.10. *Compositions of continuous maps are continuous; identities are continuous; a bijection f is a homeomorphism $\iff f^{-1}(V)$ open $\iff f(U)$ open for all opens.*

Example 1.11. $\tan$ gives $(0, 1) \cong \mathbb{R}$ (rescaled). The map $[0, 1) \rightarrow S^1$, $t \mapsto e^{2\pi it}$, is continuous and bijective but *not* a homeomorphism (its inverse is discontinuous at 1; compactness will explain this cleanly, Corollary 1.23).

Definition 1.12 (Connected). X is **disconnected** if $X = U \sqcup V$ with U, V nonempty open; otherwise **connected**.

Proposition 1.13. *The continuous image of a connected space is connected.*

Proof. If $f(X) = U \sqcup V$ is a disconnection then $X = f^{-1}(U) \sqcup f^{-1}(V)$ is a disconnection of X (preimages of opens are open, disjoint, and cover since U, V cover $f(X)$). $\square$

Theorem 1.14 (Intervals are connected). *Every interval $I \subseteq \mathbb{R}$ (in the standard topology) is connected; in particular $\mathbb{R}$ and the unit interval are connected.*

Proof. Suppose $I = U \sqcup V$ with U, V nonempty open in I . Pick $a \in U$, $b \in V$; wlog $a < b$; replace I by $[a, b] \subseteq I$ and U, V by their intersections with $[a, b]$ (still a disconnection, $a \in U$, $b \in V$). Let $s = \sup(U \cap [a, b])$; then $s \in [a, b] = U \sqcup V$.

If $s \in U$: U open in $[a, b]$ gives $\varepsilon > 0$ with $[s, s + \varepsilon) \subseteq U$ when $s < b$; then $s + \varepsilon/2 \in U$ exceeds the supremum — impossible; and $s = b$ forces $b \in U \cap V$.

If $s \in V$: V open gives $\varepsilon > 0$ with $(s - \varepsilon, s] \subseteq V$; but s being the *least* upper bound of $U \cap [a, b]$ yields $u \in U \cap (a, s] \subseteq U \cap V$ — impossible. Both cases fail, so no disconnection exists. $\square$

Example 1.15. $\mathbb{Q}$ is **totally disconnected**: for $x < y$ in $\mathbb{Q}$ pick an irrational $t \in (x, y)$; then $(-\infty, t) \cap \mathbb{Q}$ and $(t, \infty) \cap \mathbb{Q}$ disconnect any subset containing x, y . Also $[0, 1] \sqcup [2, 3]$ is disconnected. Contrast: $\mathbb{R}_{>0}$, circles, and all intervals are connected.

Definition 1.16 (Compact). X is **compact** if every open cover $\{U_i\}_{i \in I}$ of X has a finite subcover.

Example 1.17. • Finite spaces are compact; $\{1/n : n \geq 1\} \cup \{0\} \subseteq \mathbb{R}$ is compact (any cover: one open catches 0 and hence all but finitely many points).

- $(0, 1)$ is *not* compact: $(1/n, 1)_{n \geq 2}$ covers with no finite subcover.
- $\mathbb{R}$ is not compact: $(-n, n)_{n \geq 1}$ covers with no finite subcover.
- The unit interval *is* compact: Theorem 1.19.

Proposition 1.18 (Continuous images of compact sets). *The continuous image of a compact space is compact.*

Proof. If $\{V_i\}$ covers $f(X)$ then $\{f^{-1}(V_i)\}$ covers X ; a finite subcover of the latter maps to a finite subcover of the former. $\square$

Theorem 1.19 (Compactness of the unit interval). *The interval $[0, 1]$, with the standard topology, is compact.*

Proof by bisection. Assume an open cover $\mathcal{U}$ with no finite subcover. Bisect: of the two halves of $[0, 1]$, at least one cannot be finitely covered (else the union could); call it I_1 . Bisect I_1 , choose a non-finitely-coverable half I_2 ; continue, obtaining nested closed intervals I_n with $|I_n| = 2^{-n}$, none finitely covered. By the nested-interval property, $\bigcap_n I_n = \{s\}$ for a single $s \in [0, 1]$. Some $U \in \mathcal{U}$ contains s , and U open gives $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap [0, 1] \subseteq U$. Choose n with $2^{-n} < \varepsilon$; since $s \in I_n$ and $|I_n| < \varepsilon$, $I_n \subseteq (s - \varepsilon, s + \varepsilon)$, so $\{U\}$ finitely covers I_n — contradiction. $\square$

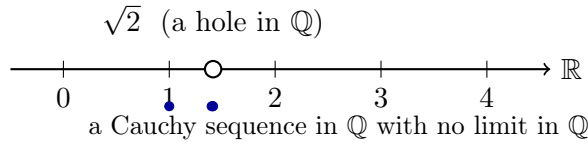


Figure 1: Incompleteness as a *missing point*: the rationals have a hole exactly where a Cauchy sequence demands a limit. Completion (§3) fills all such holes at once.

Proposition 1.20 (Closed inside compact; Heine–Borel for $\mathbb{R}$). (a) *A closed subset of a compact space is compact.*

(b) *(Heine–Borel, $n = 1$) $K \subseteq \mathbb{R}$ is compact $\iff K$ is closed and bounded.*

Proof. (a) If $K \subseteq X$ closed and $\{U_i\}$ covers K , then $\{U_i\} \cup \{X \setminus K\}$ covers X ; a finite subcover minus $X \setminus K$ finitely covers K . (b) ($\Leftarrow$) Bounded gives $K \subseteq [a, b]$; $[a, b]$ is homeomorphic to $[0, 1]$ (rescaling), hence compact (Theorem 1.19); closed-in-compact gives K compact by (a). ($\Rightarrow$) Compact K is bounded since $(-n, n)_n$ must admit a finite subcover; and K is closed by Proposition 1.22(b). $\square$

Definition 1.21 (Hausdorff). X is **Hausdorff** (T_2) if distinct points have disjoint open neighbourhoods.

Proposition 1.22 (Hausdorff consequences). (a) *Limits of sequences are unique in Hausdorff spaces.*

(b) *Compact subsets of Hausdorff spaces are closed.*

Proof. (a) If $x_n \rightarrow x$ and $x_n \rightarrow y$ with $x \neq y$, disjoint $U \ni x$, $V \ni y$ force $x_n \in U \cap V = \emptyset$ for large n . (b) Let K compact in Hausdorff X ; for $x \notin K$, each $y \in K$ has disjoint $U_y \ni x$, $V_y \ni y$; finitely many V_{y_i} cover K ; then $U = \bigcap_i U_{y_i}$ is an open neighbourhood of x disjoint from K ; so $X \setminus K$ is open. $\square$

Corollary 1.23. *Compactness is a homeomorphism invariant; hence $[0, 1] \not\cong (0, 1) \not\cong \mathbb{R}$, and the bijection $[0, 1] \rightarrow S^1$ of the example above is not a homeomorphism.*

Remark 1.24 (Zariski is not Hausdorff). In $\text{Spec } \mathbb{Z}$ (Session S1) any two nonempty opens meet: $D(m) \cap D(n) = D(mn) \neq \emptyset$. So the Zariski topology sacrifices Hausdorffness to remember algebraic specialization; the metric and étale topologies (S17–, S52) restore separation where it is needed.

2 Metric Spaces

Definition 2.1 (Metric). A **metric** on X is $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ with: $d(x, y) = 0 \iff x = y$; symmetry $d(x, y) = d(y, x)$; triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

Example 2.2. • Euclidean $d_2(x, y) = (\sum_i (x_i - y_i)^2)^{1/2}$ on $\mathbb{R}^n$; taxicab $d_1 = \sum_i |x_i - y_i|$; max metric $d_\infty = \max_i |x_i - y_i|$ (all give the standard topology on $\mathbb{R}^n$).

- Discrete metric: $d(x, y) = 1$ for $x \neq y$; induces the discrete topology.
- Sup metric on $C[0, 1]$: $d_{\text{sup}}(f, g) = \sup_{t \in [0, 1]} |f(t) - g(t)|$; convergence = uniform convergence.
- Preview (S17): the p -adic metric $d_p(x, y) = p^{-v_p(x-y)}$ on $\mathbb{Q}$; ultrametric inequality $d_p(x, z) \leq \max\{d_p(x, y), d_p(y, z)\}$ — a geometry where all triangles are isosceles (S19).

Proposition 2.3 (Balls form a basis). *The open balls $B(x, r)$ satisfy (B1), (B2); hence every metric space carries a canonical **metric topology**.*

Proof. (B1): $x \in B(x, 1)$. (B2): if $z \in B(x, r) \cap B(y, s)$ set $\varepsilon = \min\{r - d(x, z), s - d(y, z)\} > 0$; then for $w \in B(z, \varepsilon)$: $d(x, w) \leq d(x, z) + d(z, w) < d(x, z) + \varepsilon \leq r$, so $B(z, \varepsilon) \subseteq B(x, r)$; symmetrically for $B(y, s)$. $\square$

Proposition 2.4. *Every metric space is Hausdorff; hence limits of sequences are unique.*

Proof. For $x \neq y$, $r = d(x, y)/2$ gives disjoint $B(x, r)$, $B(y, r)$ (triangle inequality). Then apply Proposition 1.22(a). $\square$

Definition 2.5 (Convergence; Cauchy; completeness). $x_n \rightarrow x$ means $\forall \varepsilon > 0 \exists N \forall n \geq N : d(x_n, x) < \varepsilon$. (x_n) is **Cauchy** if $\forall \varepsilon > 0 \exists N \forall m, n \geq N : d(x_m, x_n) < \varepsilon$. (X, d) is **complete** if every Cauchy sequence converges in X .

Proposition 2.6. *Every convergent sequence is Cauchy.*

Proof. Given ε , choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$; then $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon$. $\square$

Example 2.7 (Incompleteness as missing points). In $\mathbb{Q}$: the decimals $1, 1.4, 1.41, 1.414, \dots$ of $\sqrt{2}$ are Cauchy (they converge in $\mathbb{R}$, use Proposition 2.6) but have no limit in $\mathbb{Q}$ (no rational squares to 2). In $(0, 1)$: $x_n = 1/n$ is Cauchy with no limit in the space. In both cases a *point is missing* (Figure 1).

Theorem 2.8 (Completeness of the sup metric). *The metric space $(C[0, 1], d_{\sup})$ is complete: every Cauchy sequence of continuous functions converges uniformly to a continuous function.*

Proof. Let (f_n) be Cauchy. For each t , $(f_n(t))$ is Cauchy in $\mathbb{R}$ (since $|f_m(t) - f_n(t)| \leq d_{\sup}(f_m, f_n)$), hence converges; define $f(t) = \lim_n f_n(t)$. Uniformity: given ε , choose N with $d_{\sup}(f_m, f_n) < \varepsilon$ for $m, n \geq N$; fixing $n \geq N$ and letting $m \rightarrow \infty$ gives $|f(t) - f_n(t)| \leq \varepsilon$ for all t , so $d_{\sup}(f, f_n) \leq \varepsilon$: $f_n \rightarrow f$ uniformly. Continuity of f : for t_0 and ε , pick n with $d_{\sup}(f, f_n) < \varepsilon/3$, then δ with $|f_n(t) - f_n(t_0)| < \varepsilon/3$ for $|t - t_0| < \delta$; then $|f(t) - f(t_0)| \leq |f(t) - f_n(t)| + |f_n(t) - f_n(t_0)| + |f_n(t_0) - f(t_0)| < \varepsilon$. $\square$

Proposition 2.9. *In a complete metric space, a subspace is complete $\iff$ it is closed. In particular complete subspaces of complete spaces are closed, and closed subspaces of complete spaces are complete.*

Proof. If Y complete and $x_n \rightarrow x$ with $x_n \in Y$, then (x_n) Cauchy in Y , so converges in Y to some $y \in Y$; uniqueness of limits gives $x = y \in Y$: Y closed. Conversely if Y closed and (x_n) Cauchy in Y , it converges in X to some x ; closedness forces $x \in Y$: convergence in Y . $\square$

Theorem 2.10 (Epsilon–delta versus open sets). *For $f : X \rightarrow Y$ between metric spaces, the following are equivalent:*

(a) *f is continuous in the open-set sense;*

(b) *f is continuous at every point: $\forall x \forall \varepsilon > 0 \exists \delta > 0 : d_X(x', x) < \delta \Rightarrow d_Y(f(x'), f(x)) < \varepsilon$.*

Proof. (a) $\Rightarrow$ (b): $B(f(x), \varepsilon)$ is open, so $f^{-1}(B(f(x), \varepsilon))$ is open and contains x ; hence it contains some $B(x, \delta)$, which is exactly (b). (b) $\Rightarrow$ (a): let V open and $x \in f^{-1}(V)$; then $f(x) \in V$ so $B(f(x), \varepsilon) \subseteq V$ for some ε ; (b) gives δ with $f(B(x, \delta)) \subseteq B(f(x), \varepsilon) \subseteq V$; thus each point of $f^{-1}(V)$ has a ball inside $f^{-1}(V)$: open by Proposition 1.4. $\square$

3 Completion: Adding the Missing Points

Definition 3.1 (Completion). A **completion** of (X, d) is a complete metric space $(\hat{X}, \hat{d})$ together with an isometric embedding $\iota : X \rightarrow \hat{X}$ whose image is dense in $\hat{X}$.

Theorem 3.2 (Existence of completions). *Every metric space has a completion.*

Proof. Let $\mathcal{C}$ be the set of Cauchy sequences in X ; declare $(x_n) \sim (y_n)$ iff $d(x_n, y_n) \rightarrow 0$ (an equivalence relation). Set $\hat{X} = \mathcal{C}/\sim$ and

$$\hat{d}([(x_n)], [(y_n)]) = \lim_n d(x_n, y_n).$$

The limit exists: $|d(x_m, y_m) - d(x_n, y_n)| \leq d(x_m, x_n) + d(y_m, y_n) \rightarrow 0$, so $(d(x_n, y_n))$ is Cauchy in $\mathbb{R}$, hence convergent ($\mathbb{R}$ complete). *Independent of representatives:* if $(x'_n) \sim (x_n)$ and $(y'_n) \sim (y_n)$ then $|d(x'_n, y'_n) - d(x_n, y_n)| \leq d(x'_n, x_n) + d(y'_n, y_n) \rightarrow 0$. *Metric axioms:* symmetry and triangle pass to the limit; $\hat{d} = 0 \iff d(x_n, y_n) \rightarrow 0 \iff (x_n) \sim (y_n)$. *Embedding:* $\iota(x) = [(x, x, \dots)]$ satisfies $\hat{d}(\iota(x), \iota(y)) = d(x, y)$. *Density:* for $[(x_n)]$ and ε , choose N with $d(x_m, x_n) < \varepsilon$ for $m, n \geq N$; then $\hat{d}(\iota(x_N), [(x_n)]) = \lim_m d(x_N, x_m) \leq \varepsilon$. *Completeness:* let $([\xi^{(k)}])_k$ be Cauchy in $\hat{X}$; by density pick $x_k \in X$ with $\hat{d}(\iota(x_k), \xi^{(k)}) < 1/k$; then (x_k) is Cauchy in X (triangle inequality), and $\hat{d}(\iota(x_k), \xi^{(k)}) \rightarrow 0$ forces $\iota(x_k) \rightarrow$ the given Cauchy class; hence $[(x_k)] \in \hat{X}$ is the limit. So $\hat{X}$ is complete. $\square$

Theorem 3.3 (Uniqueness of completions). *Any two completions $(\hat{X}_1, \iota_1)$, $(\hat{X}_2, \iota_2)$ of (X, d) are related by a unique isometry $\Phi : \hat{X}_1 \rightarrow \hat{X}_2$ with $\Phi \circ \iota_1 = \iota_2$.*

Proof idea. The map $\iota_2 \circ \iota_1^{-1} : \iota_1(X) \rightarrow \iota_2(X)$ is an isometry between dense subspaces; isometries are uniformly continuous, and uniformly continuous maps from a dense subspace into a *complete* space extend uniquely continuously; the extensions of Φ and of its inverse are mutually inverse isometries by uniqueness of continuous extensions from dense subspaces. $\square$

Example 3.4 (Completions we will live with). • The completion of $(\mathbb{Q}, |\cdot|)$ is $\mathbb{R}$ (Cantor's construction is exactly Theorem 3.2).

- The completion of $((0, 1), |\cdot|)$ is $[0, 1]$.
- The completion of $(\mathbb{Q}, d_p)$ is $\mathbb{Q}_p$, and $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/p^n$ appears in Sessions S17–S18: the same machine, a different metric, a different world.
- $(C[0, 1], d_{\sup})$ is already complete (Theorem 2.8); its L^2 -metric completion is the Lebesgue space $L^2[0, 1]$ (mention; not used later).

Remark 3.5 (Ostrowski preview: the census of completions of $\mathbb{Q}$). Ostrowski's theorem (Session S17) says that, up to equivalence, the only non-trivial metrics on $\mathbb{Q}$ are $|\cdot|$ and the d_p ; hence the completions of $\mathbb{Q}$ are exactly $\mathbb{R}$ and the $\mathbb{Q}_p$. The course's slogan: *each completion adds precisely the points its Cauchy sequences deserved; the adeles (S22) then gather all completions into one space.*

4 Compactness and Continuity in Metric Spaces

Definition 4.1 (Sequential compactness; total boundedness). X is **sequentially compact** if every sequence has a convergent subsequence. X is **totally bounded** if for every $\varepsilon > 0$ finitely many ε -balls cover X .

Proposition 4.2. *For metric spaces: compact $\Rightarrow$ sequentially compact $\Rightarrow$ complete and totally bounded; and complete + totally bounded $\Rightarrow$ compact. Hence in metric spaces all three notions coincide.*

Proof ideas. Compact $\Rightarrow$ seq. compact: if some sequence had no convergent subsequence, each $x \in X$ would have a neighbourhood U_x meeting the sequence in finitely many terms; finitely many U_x cover X , yet the sequence has infinitely many terms — contradiction. Seq. compact $\Rightarrow$ totally bounded: else some $\varepsilon > 0$ admits an infinite ε -separated sequence, which has no Cauchy (hence no convergent) subsequence. Seq. compact $\Rightarrow$ complete: a Cauchy sequence with a convergent subsequence converges to the same limit. Complete + totally bounded $\Rightarrow$ compact: from any open cover build, via total boundedness and a Lebesgue-number argument, a finite subcover (standard; details in Munkres §32, Sutherland ch.6). $\square$

Theorem 4.3 (Uniform continuity on compact metric spaces). *A continuous map from a compact metric space to a metric space is uniformly continuous: $\forall \varepsilon \exists \delta \forall x, x' : d_X(x, x') < \delta \Rightarrow d_Y(f(x), f(x')) < \varepsilon$.*

Proof. Given ε , for each x choose δ_x with $f(B(x, \delta_x)) \subseteq B(f(x), \varepsilon/2)$. The balls $B(x, \delta_x/2)$ cover X ; compactness gives finitely many $B(x_i, \delta_{x_i}/2)$; set $\delta = \min_i \delta_{x_i}/2 > 0$. If $d(x, x') < \delta$, pick i with $x \in B(x_i, \delta_{x_i}/2)$; then $d(x', x_i) \leq d(x', x) + d(x, x_i) < \delta + \delta_{x_i}/2 \leq \delta_{x_i}$; hence both $f(x), f(x')$ lie in $B(f(x_i), \varepsilon/2)$, so $d_Y(f(x), f(x')) < \varepsilon$. $\square$

Corollary 4.4 (Extreme values). *A continuous $f : X \rightarrow \mathbb{R}$ on a nonempty compact space attains its maximum and minimum.*

Proof. $f(X)$ is compact in $\mathbb{R}$ (Proposition 1.18), hence closed and bounded (Proposition 1.20); so $\sup f(X), \inf f(X) \in f(X)$. $\square$

Remark 4.5 (What this session hands to the course). Compactness will reappear as: compactness of $\mathbb{Z}_p$ (S19) and of profinite groups (S34); adelic compactness $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ (S22), the engine of class-group finiteness (S24); quasi-compactness of Spec (S1, S33); compact Riemann surfaces (S38). Completion will reappear as $\mathbb{Q}_p$ (S17–S18), $k((t))$ (S21), and profinite completions of groups (S34).

5 Exercises

Exercise 1. Verify that the cofinite collection on an infinite set satisfies (T1)–(T3), and that it is not Hausdorff. Which sequences converge, and to how many limits?

Exercise 2. Show that the collection $\{(a, b] : a < b\}$ is a basis for a topology on $\mathbb{R}$ (the lower-limit topology); prove it is strictly finer than the standard topology and Hausdorff, and that $[0, 1)$ is open in it.

Exercise 3. Prove that a subset of a subspace is open in the subspace topology $\iff$ it is the intersection of the subspace with an open set of the ambient space; deduce that openness is not transitive (give an example).

Exercise 4. Prove Proposition 1.13 and deduce that $[0, 1]$ and $[0, 1) \sqcup (1, 2]$ are not homeomorphic.

Exercise 5. Show $\{1/n : n \geq 1\} \cup \{0\}$ is compact directly from the definition, and that $\{1/n : n \geq 1\}$ (without 0) is not.

Exercise 6. Prove that the discrete metric induces the discrete topology, and that a metric space is discrete $\iff$ every subset is closed $\iff$ every function from it is continuous.

Exercise 7. Prove Proposition 2.3 in full detail, and show d_1, d_2, d_{∞} on $\mathbb{R}^n$ all generate the standard topology (compare balls: $B_{\infty} \subseteq B_2 \subseteq B_1 \subseteq n B_{\infty}$).

Exercise 8. In $(\mathbb{Q}, |\cdot|)$, prove directly (without invoking $\mathbb{R}$) that the sequence $x_1 = 1$, $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ is Cauchy and has no limit in $\mathbb{Q}$.

Exercise 9. Carry out Theorem 3.2 for $X = (0, 1)$ explicitly: identify $\hat{X}$ with $[0, 1]$ by describing the equivalence classes of Cauchy sequences that correspond to 0 and 1.

Exercise 10. Prove that a closed subset of a complete metric space is complete, and that $\mathbb{Z} \subseteq \mathbb{R}$ and $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ are closed (the latter using the ultrametric inequality, preview of S19).

Exercise 11. Show that the product topology on $\mathbb{R}^{\mathbb{N}}$ (countable product) is strictly coarser than the box topology, by exhibiting a box-open set that contains no product-basic neighbourhood of the origin.

Exercise 12. Prove Theorem 4.3 again by contradiction using sequential compactness, and deduce that every continuous $f : S^1 \rightarrow \mathbb{R}$ is uniformly continuous.

Where This Session Feeds the Course

- **Depends on:** nothing (entry point); P1 supplies the group language used in examples.
- **Feeds:** S17–S18 (completion $\mathbb{Q} \rightarrow \mathbb{Q}_p$; ultrametric topology); S19 (compactness of $\mathbb{Z}_p$); S21 ($k((t))$ as completion); S22 (restricted product, adelic compactness — the finiteness engine for S24); S33–S34 (quasi-compactness, profinite compactness); S37–S38 (complex analysis, compact Riemann surfaces); S52 (contrast with Zariski/étale topologies).
- **Unifying thread:** completion adds exactly the points Cauchy sequences deserved — $\mathbb{R}$ here, $\mathbb{Q}_p$ in Act III; compactness is what makes infinite point-counts finite, from the unit interval to $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$.

References for this session

- [1] Munkres, J.R. *Topology*, ch.2–4 (bases, subspace/product, connectedness, compactness).
- [2] Sutherland, W.A. *Introduction to Metric and Topological Spaces* (metric viewpoint).
- [3] Rudin, W. *Principles of Mathematical Analysis*, ch.2–4 (completion; completeness of $C[0, 1]$).
- [4] Gouvêa, F.Q. *p-adic Numbers: An Introduction*, ch.1–3 (the p -adic preview).

End of Session P2 — next: Session P3, Algebraic Topology Crash Course